

SALES CENTER MODULE

Complete System Documentation

Global ERP System Date: March 29, 2026 Version: 1.0

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1. Executive Summary

The Sales Center module is a comprehensive telephony and CRM platform integrated into the Global ERP system. It provides end-to-end call management, agent performance tracking, AI-powered insights, and automated lead generation.

Key Capabilities:

- Yeastar PBX integration via Linkus SDK for browser-based calling
 - Real-time call tracking with webhook-driven status updates
 - AI-powered call analysis with automatic lead creation
 - Agent performance scoring with configurable thresholds
 - Inhouse vs Remote team comparison dashboards
 - Three AI intelligence engines (e2, e7, e8) generating diagnostic, predictive, and prescriptive signals
 - Customer assignment and segmentation (CHSC, Non-CHSC, Insurance)
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2. Call Management (Inbound/Outbound)

2.1 Inbound Call Flow

- 1. Customer dials company number
- 2. Yeastar PBX routes call to agent extension
- 3. PBX sends webhook to `POST /api/webhooks/dialer/yeastar`

4. System creates **call_session** record with status **ringing**
5. Agent answers -> PBX sends status update -> status becomes **in_progress**
6. Call ends -> PBX sends CDR webhook -> status becomes **completed** with duration
7. System matches caller number to existing customer (if found)
8. AI workflow evaluates call for lead creation eligibility

2.2 Outbound Call Flow

1. Agent clicks call button on customer card (Agent Dashboard) or uses Outbound Dialer
2. Browser Linkus SDK initiates call via PBX
3. System creates **call_session** with direction **outbound**
4. PBX routes call through configured outbound trunk
5. Webhook updates track call progress (ringing -> in_progress -> completed)
6. Call recording is captured and stored

2.3 Call Statuses

Status	Description	Color
initiated	Call creation requested	Blue
ringing	Ringling on recipient device	Amber
in_progress	Call connected and active	Amber
completed	Call successfully ended	Green
failed	Call failed to connect	Red
cancelled	Call cancelled before completion	Red

2.4 Yeastar SDK Integration

- **Connection:** Linkus SDK connects to PBX via **carguru.ras.yeastar.com**
- **Authentication:** API credentials (accessId, accessKey, username, password)
- **Extension Assignment:** Each agent has a PBX extension stored in **users.mobile**
- **Webhook Events:** 30012 (Call End Details Notification), 30020 (uaCSTA Call Report)
- **Auto Answer:** Configurable per integration

3. Call History & Recordings

3.1 Call History Page

Route: **/company/{companyId}/call-center/history**

Features:

- Direction filter: All / Inbound / Outbound
- KPI bar: Total Calls, Inbound, Outbound, Completed, Missed, Avg Duration, Answer Rate
- Sortable columns: Customer, Direction, From, To, Status, Started, Agent
- Inline notes: Add/edit notes per call with save button

- Recording playback: HTML5 audio player with download
- "Resolve Recording" button to fetch recording from PBX if not yet available

3.2 AI Inquiries Tab

Displays calls analyzed by AI for lead generation potential:

- Filters: All / Not Converted / Converted
- Shows: Customer match, verification status, AI analysis confidence, lead type
- Actions: Verify, Analyze (AI), Convert to Lead, Set Lead Type

3.3 Call Recordings

- Recordings captured automatically by Yeastar PBX
- Stored with call session reference
- Accessible via authenticated proxy endpoint
- Duration tracking for quality monitoring

4. Agent Management

4.1 User Extensions Page

Route: `/company/{companyId}/integrations/dialer/extensions`

- Assign PBX extension numbers to agents (stored in `users.mobile`)
- Set agent location: **Inhouse** or **Remote** (stored in `users.dialer_location`)
- View all company users with their extension status
- Extension used for call routing and agent identification

4.2 Agent Roles & Permissions

Role	Access Level
Agent	Own dashboard, assigned customers, make calls
Supervisor	Agent reports, Data Center, team KPIs
Company Admin	Full access: settings, thresholds, all agents

4.3 Agent Location Tracking

- `inhouse`: Agents working from office
- `remote`: Agents working from home
- Used for team split in Master Dashboard and AI comparison signals
- Default: `inhouse` if not set

5. Agent Dashboard

Route: `/company/{companyId}/call-center/agent-dashboard`

5.1 Header

- Agent avatar and name
- Agent selector dropdown (visible to managers/admins only)
- Refresh button
- AI Insights toggle button

5.2 KPI Cards (6 metrics)

KPI	Description	Color Coding
Assigned	Active customers assigned to agent	Blue
Contacted	Customers agent has called	Green
Pending	Customers awaiting follow-up	Amber
Total Calls	All calls made/received	White
Answer Rate	Completed / Total calls	Green >= 80%, Amber >= 50%, Red < 50%
Avg Duration	Average call length	Gray

5.3 Call History Tab

- Relative timestamps ("just now", "5m ago", "2h ago")
- Direction icons: Inbound (green down arrow), Outbound (blue up arrow)
- Phone numbers in monospace font
- Customer name linked (if matched)
- Status badges with color coding
- Duration in MM:SS format

5.4 Customers Tab

- **Search:** Real-time filter by name, phone, email
- **Segment Filter:** All / CHSC / Non-CHSC / Insurance (toggle buttons)
- **Customer Name:** Clickable link to customer dashboard (blue text)
- **Call Button:** Phone icon next to each number - click to dial directly
- **Segment Badge:** Color-coded pill (CHSC: purple, Insurance: blue, Non-CHSC: gray)
- **Assigned Date:** Formatted date

5.5 AI Coaching Sidebar (Right Panel)

- Fixed 420px sidebar on desktop (bottom sheet on mobile)
- Engines: e2 (Agent Performance) + e7 (Coaching Intelligence)
- Shows Diagnostic, Predictive, Prescriptive signals
- Auto-refreshes every 5 minutes
- Signal cards with urgency badges (HIGH: red, MED: amber, LOW: green)

5.6 Access Control

- Agents see only their own data (locked to own userId)

- Supervisors can view their team's agents
- Admins can select any agent from dropdown

6. Supervisor Performance Summary

Route: `/company/{companyId}/call-center/performance`

6.1 Header

- Title: "Sales Center Performance"
- Agent count display
- Date range picker (from/to datetime-local)
- "Today" quick-reset button
- Refresh button
- AI Insights toggle

6.2 KPI Cards (6 cards with delta comparison)

KPI	Icon	Delta	Color Logic
Total Calls	Phone	vs yesterday %	White
Answer Rate	Check	vs yesterday %	Green/Amber/Red thresholds
Avg Duration	Clock	vs yesterday %	White
Collection	Currency	vs yesterday %	White
Appointments	Calendar	vs yesterday %	White
CHSC Conv.	Target	Target line	Green if >= target, else Amber

Delta arrows: Green up arrow for positive, Red down arrow for negative.

6.3 Agent Rankings Table

Sorted by performance score (descending):

- **Rank:** Sequential number
- **Agent Name:** Full name or extension
- **Score:** Visual progress bar (0-100) with color gradient
- **Calls / Held / Held% / Miss%:** Call volume metrics
- **Avg Duration:** MM:SS format
- **CHSC Sold / CHSC In / Conv%:** CHSC conversion metrics
- **Non-CHSC / Appt Today / Appt Tmrw:** Additional metrics
- **Collection / Rev/Call:** Revenue metrics
- **Badge:** Performance badge with dot indicator

6.4 Performance Badges

Badge	Score	Color
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Badge	Score	Color
Excellent	>= 85	Emerald
Good	>= 70	Cyan
Average	>= 50	Amber
Needs Improvement	< 50	Rose

6.5 Performance Score Formula

Score = (Held Rate % x 40 + CHSC Conv % x 25 + Rev/Call normalized x 20 +
Appointments normalized x 15) / 100

All weights configurable via admin settings.

6.6 AI Intelligence Sidebar

- Fixed right panel (420px) with e8 engine
- Shows Diagnostic, Predictive, Prescriptive tabs
- Compares inhouse vs remote agent groups
- Signal examples:
 - "Agent X held rate 15pp below average - schedule coaching"
 - "CHSC conversion trending down 20% - review scripts"
 - "Remote team outperforming inhouse by 10% in answer rate"

7. Master Performance Dashboard

Route: /company/{companyId}/master-dashboard

7.1 Top 5 KPI Cards

KPI	Format	Comparison
Total Revenue	AED currency	vs previous period %
Total Bookings	Count	vs previous period %
Total Calls Handled	Count	vs previous period %
Show-Up Rate	Percentage	vs previous period %
Total Cancellations	Count	vs previous period %

7.2 Team Performance Summary

Three team rows in table:

- **Inhouse Team:** Office-based agents with KPIs

- **Remote Team:** Remote agents with KPIs
- **Portfolio Team:** Account management metrics

Each row shows: Total Calls, Answer Rate, Bookings, Conversion Rate, Score, Grade (A+ to D)

7.3 Inhouse vs Remote Team (Side-by-Side)

Two columns displayed simultaneously for easy comparison:

Per Team (Inhouse/Remote):

- 4 KPI cards: Total Calls, Answer Rate, Avg Duration, Abandonment Rate
- Call Trend sparkline chart
- 3 KPI rows with progress bars: Answer Rate, Abandonment, Score
- Status badges: Excellent/Good/Warning/Poor

7.4 Conversion Funnel

Visual funnel showing: Total Calls -> Connected -> Leads Created -> Bookings -> Show-Ups -> Revenue
Overall conversion rate displayed.

7.5 Portfolio Team Section

- 6 KPI cards: Active Accounts, Monthly Touchpoints, Return Bookings, Return Revenue, Upsell Rate, Retention Rate
- Revenue Breakdown pie chart (Workshop / RSA / Recovery / Other)
- Top Performers table (MTD)
- Needs Attention alerts

7.6 AI Intelligence Sidebar

- e8 engine with inhouse vs remote comparison
- Diagnostic signals for held rate gaps
- Predictive signals for collection trajectory
- Prescriptive signals for coaching actions

8. Lead Creation & AI Workflow

8.1 AI-Powered Lead Creation (Automatic)

1. Inbound call received and recorded
2. AI analyzes call recording (OpenAI/Anthropic)
3. Extracts: caller intent, vehicle plate, service type, location
4. Creates `call_ai_inquiry` record with analysis results
5. If `auto_convert_eligible = true` and policy mode is `live`:
 - Automatically creates lead with extracted data
 - Sets `conversion_status = 'converted'`
 - Links to matched customer (if found by phone number)

8.2 Manual Lead Creation from CRM

1. Agent views Call History or AI Inquiries tab
2. Clicks "Convert to Lead" on an inquiry
3. System creates lead with pre-populated data:
 - Customer (matched by phone or new)
 - Lead type: RSA / Recovery / Workshop
 - Source: Call inquiry
 - Car plate (if extracted by AI)
4. Agent can also click "Create Lead" from any call in history
5. Redirects to lead creation form with pre-filled phone/customer data

8.3 AI Inquiry Analysis

When "Analyze (AI)" is clicked:

- Fetches call recording from PBX
- Sends to AI for transcription and analysis
- Extracts: lead type, request type, plate number, location, caller name, mobile, summary
- Calculates confidence score (0-100%)
- Determines auto-convert eligibility

8.4 Call AI Policy

Configurable per company:

- **dry_run mode:** AI analyzes but doesn't auto-convert (logging only)
- **live mode:** AI auto-converts eligible inquiries to leads
- **Business hours:** Restrict AI processing to configured hours
- **Restrictions:** Block unknown numbers, blocked/allowed prefixes
- **Guidance:** Welcome message, system prompt, escalation keywords

9. Data Center & Customer Assignment

Route: `/company/{companyId}/data-center`

9.1 Auto-Assign Customers

- Select agent from dropdown (active agents only)
- Set total customers to assign (default: 100)
- Configure distribution percentages:

Segment	Description
CHSC Active	Active CHSC customers
CHSC Inactive	Inactive CHSC customers
Non-CHSC Active	Active non-CHSC customers
Non-CHSC Inactive	Inactive non-CHSC customers
Insurance	Insurance customers

- Total must equal 100% (live validation)
- "Set Equal (20%)" quick button
- Results show: assigned count per segment

9.2 Agent Performance Report

Table showing all agents with:

- Total Assigned, Total Calls, Answered, Failed
- Call Duration, Total Leads, Total Collection
- Answer Rate %, Performance Score, Performance Label

9.3 KPIs per Agent

- Assigned Customers, Contacted, Pending
- Total Calls, Answered, Failed
- Answer Rate %, Avg Call Duration

9.4 Export

- Excel and PDF export for agent reports
- Filtered by date range and segment

10. AI Intelligence Layer

10.1 Engine Architecture

The system uses Anthropic Claude Sonnet 4.6 with 8 specialized engines:

Engine	Name	Used In
e1	Funnel Intelligence	Lead pipeline analysis
e2	Agent Performance	Agent Dashboard sidebar
e3	Revenue Forecasting	Revenue predictions
e4	Churn & Retention	Customer risk scoring
e5	Anomaly Detection	Statistical anomaly alerts
e6	Collections Intelligence	Overdue invoice ranking
e7	Coaching Intelligence	Agent Dashboard sidebar
e8	Sales Center Performance Intelligence	Performance Summary + Master Dashboard

10.2 Signal Types

- **Diagnostic:** Root cause analysis ("Why is held rate dropping?")
- **Predictive:** Trend forecasting ("Collection will decline 15% next week")
- **Prescriptive:** Action recommendations ("Schedule coaching for Agent X within 48h")

10.3 Signal Fields

Each signal includes: type, metric, observation, diagnosis, action, urgency (HIGH/MED/LOW), owner_role, respond_by_hours, confidence (0.0-1.0), expected_outcome

10.4 e8 Engine: Sales Center Performance Intelligence

Focus areas:

- Agents with held rate significantly below company average
- CHSC conversion imbalances
- Agents with high call volume but zero appointments
- Top performers to recognise
- Collection-per-call outliers
- **Inhouse vs Remote comparison:** Flags gaps between groups

Data payload includes:

- Per-agent: total_calls, held_calls, held_rate_pct, held_rate_delta_pct, chsc metrics, collection, appointments
- Group summaries: inhouse_summary, remote_summary (agents count, held rate, total calls)

10.5 Caching & Refresh

- 5-minute in-memory cache per company/engine
- Configurable refresh interval (1-60 min) per engine
- Cache invalidated on config update

10.6 Admin Configuration

Route: /company/{companyId}/settings/call-center/performance

Configurable Thresholds:

Threshold	Default	Controls
badge_excellent_min	85	Excellent badge cutoff
badge_good_min	70	Good badge cutoff
badge_average_min	50	Average badge cutoff
held_rate_green_min	80	Green color threshold
held_rate_amber_min	50	Amber color threshold
weight_held_rate	40	Score weight %
weight_chsc_conversion	25	Score weight %
weight_revenue_per_call	20	Score weight %
weight_appointments	15	Score weight %

Threshold	Default	Controls
target_chsc_loyalty_pct	40	KPI target line
target_appointment_show_pct	70	KPI target
target_callback_rate_pct	85	KPI target
alert_held_rate_gap_pp	15	AI alert threshold

11. Customer List

Route: `/company/{companyId}/call-center/customers-list`

Tabs

Tab	Filter	Extra Features
CHSC	customer_type = 'CHSC'	-
Non-CHSC	All non-CHSC/Insurance types	-
Insurance	Insurance type customers	Policy expiry tracking, expiry status filter

Features

- Search by name, phone, email, code
- Sort by created_at or name (asc/desc)
- Pagination: 10/25/50/100 per page
- Customer name clickable to profile
- Insurance: Expiry status badges (Active: green, Expire Soon: amber, Expired: red)

12. API Reference

Call Management

Method	Endpoint	Description
POST	<code>/api/webhooks/dialer/yeastar</code>	Yeastar webhook receiver
POST	<code>/api/company/{id}/call-center/call</code>	Place outbound call
GET	<code>/api/company/{id}/call-center/history</code>	Call history with filters
GET	<code>/api/company/{id}/call-center/sessions</code>	Recent call sessions
GET	<code>/api/company/{id}/call-center/active</code>	Active/live calls

Agent & Performance

Method	Endpoint	Description
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Method	Endpoint	Description
GET	/api/company/{id}/call-center/agent-summary	Agent performance summary
GET	/api/company/{id}/call-center/agent-dashboard/calls	Agent's call history
GET	/api/company/{id}/call-center/agent-dashboard/customers	Agent's assigned customers
GET	/api/company/{id}/master-dashboard	Master dashboard data

Data Center

Method	Endpoint	Description
GET	/api/company/{id}/data-center/kpis	KPIs by segment/agent
GET	/api/company/{id}/data-center/users	Agent list
GET	/api/company/{id}/data-center/reports/agents	Agent performance report
POST	/api/company/{id}/data-center/assignments/auto	Auto-assign customers

AI Intelligence

Method	Endpoint	Description
GET	/api/company/{id}/intelligence/signals	Fetch AI signals
GET	/api/company/{id}/intelligence/config	Get engine configs
PATCH	/api/company/{id}/intelligence/config	Update thresholds

AI Provider Configuration

Method	Endpoint	Description
GET	/api/company/{id}/ai/provider	Get provider configs (OpenAI + Anthropic)
POST	/api/company/{id}/ai/provider	Save provider config

13. Configuration & Settings

13.1 AI Provider Setup

Route: /company/{companyId}/settings/ai/config

- **OpenAI:** Base URL, API Key, Active toggle
- **Anthropic (Claude):** API Key, Active toggle (required for AI Intelligence engines)
- **Linkus SDK:** PBX Server URL, Default Extension

13.2 Performance Thresholds

Route: `/company/{companyId}/settings/call-center/performance`

- Badge thresholds, Held rate colors, Score weights (must sum to 100%), KPI targets

13.3 AI Intelligence Settings

Route: `/company/{companyId}/settings/ai/intelligence`

- Per-engine: Enable/disable, Refresh interval, Custom prompt override, Threshold JSON

13.4 Sidebar Navigation

Under "Sales Center" section:

- Call History
- Customers List
- Data Center
- User Extensions
- Master Performance
- Performance Summary
- Agent Dashboard

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